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1. **Borup Fiord Pass, Ellesmere Island**
   Scope Definition: The supraglacial spring system, including the active glacial discharge outlets, associated elemental sulfur and mineral deposits (aufeis), and seasonal surface melt pools.

   a. Temperature (°C): -0.3 to 2.3
      Source: Most comprehensive field measurements state that the spring discharge waters are low, typically ranging from 1 to 2.3 °C (Grasby et al., 2003). Other subsequent sampling expeditions recorded slightly colder temperatures at the spring outlets, measuring between -0.2 and -0.3 °C (Gleeson et al., 2012, Wright et al., 2013). This minor disagreement reflects natural seasonal or inter-annual variations in glacial meltwater mixing and ambient air temperatures, as measurements were taken during different summer field campaigns.

      Reliability: 3 (High). Temperature data derives from direct, in-situ field measurements taken using calibrated multiparameter meters across multiple peer-reviewed studies.

   b. pH: 5.5 to 9.5
      Source: Field measurements indicate the primary spring discharge fluids are distinctly alkaline, displaying pH values typically between 9.0 and 9.5 (Grasby et al., 2003). Conversely, other observations land on lower, circumneutral to slightly acidic values ranging from 5.5 to 7.8 for various springs and supraglacial melt pools (Gleeson et al., 2012, Trivedi et al., 2018). The nature of this disagreement stems from varying degrees of dilution by slightly acidic glacial meltwater and the sampling of different localized sub-environments over distinct years.

      Reliability: 3 (High). Multiple peer-reviewed astrobiological and geochemical studies utilized calibrated field probes and test strips to directly measure the fluid.

   c. Salinity (g/kg): 0.1 to 10.0
      Source: Early expeditions by Grasby et al. (2003) say the total dissolved solids load ranges from 107 to 183 mg/L because these samples were heavily diluted by supraglacial meltwater during the late summer. Conversely, a later group reporting on earlier season sampling says the fluids are much more saline, measuring total dissolved solids values of 5900 mg/L up to 10,000 mg/L, identifying chloride concentrations as high as 68.8 mM (Gleeson et al., 2012, Trivedi et al., 2018). The Planetary Society blog also noted values representing about one-third the salinity of seawater. A union range of 100 to 10,000 mg/L (0.1 to 10 g/kg) is adopted because salinity fluctuates drastically based on the season, subglacial evaporite dissolution, and the degree of surface glacial meltwater mixing.

      Reliability: 3 (High). Underlying ionic concentrations were quantified using standard laboratory ion chromatography and mass spectrometry on preserved samples.

d. Pressure (MPa): 0.1
Source: Geochemical models characterize the supraglacial spring environment at a standard atmospheric pressure of roughly 1 bar, or 0.1 MPa (Wright et al., 2013). Other sources inherently agree with this surface pressure, as they describe the springs openly discharging directly onto the surface of the glacial ice and forming exposed pools (Grasby et al., 2003, Trivedi et al., 2018). There is no disagreement among the sources regarding the surface pressure of these exposed supraglacial discharge sites.

Reliability: 3 (High). High reliability is given to this parameter because it is a fundamental, verified geographic fact that the exposed supraglacial springs exist at standard atmospheric sea-level pressure.

e. Isolation Score (1-5): 2
(i) Physical Confinement (1-5): 2
Source: Grasby et al. (2003) describe the springs discharging directly onto the flat glacial surface or into meltwater channels. The supraglacial system is only partially confined, featuring seasonal ice cover, aufeis accumulation, and direct exposure to the open atmosphere.

(ii) Hydrologic Connectivity (1-5): 2
Source: The supraglacial sulfur springs exhibit seasonal exchange, heavily interacting and mixing with surrounding glacial meltwater streams and pools during the summer thaw (Grasby et al., 2003, Trivedi et al., 2018). This demonstrates connectivity with a few local, dynamic hydrological bodies.

(iii) Isolation Time (1-5): 2
Source: Subglacial fluids discharge dynamically to the surface where they freeze into aufeis in the winter and melt into pools during the summer (Lau et al., 2017). This rapid, seasonally driven physical turnover indicates an isolation time well under 1000 years.

Reliability 3 (High). The springs' physical exposure and seasonal hydrological mixing were directly observed and consistently documented in multiple peer-reviewed field studies (Grasby et al., 2003, Trivedi et al., 2018).

f. Redox Score (1-5): 4.33
(i) Redox Species (1-5): 4
Source: The supraglacial spring waters simultaneously contain sulfur in three distinct oxidation states, pairing high concentrations of reduced hydrogen sulfide with oxidized sulfate and elemental sulfur alongside atmospheric oxygen (Grasby et al., 2003, Gleeson et al., 2012). This represents strong evidence for both oxidants and reductants.

(ii) Equilibrium (1-5): 4
Source: Free energy calculations confirm that the aerobic oxidation of elemental sulfur and other reduced sulfur species is highly exergonic and energetically favorable (Wright et al., 2013). This provides strong thermodynamic evidence for an exploitable chemical gradient far from equilibrium.

(iii) Chemolithotrophic Metabolism (1-5): 5
Source: Metagenomic and 16S rRNA gene sequencing data demonstrate that the ecosystem is dominated by Epsilonproteobacteria, specifically the chemolithoautotrophic sulfur-oxidizers Sulfurovum and Sulfuricurvum (Wright et al., 2013, Trivedi et al., 2018). This constitutes strong, confirmed evidence of active chemolithotrophic metabolism.

Reliability 3 (High). High reliability is awarded because the redox scores are backed by peer-reviewed metagenomic sequencing, extensive free energy modeling, and precise stable isotope geochemistry.

2. **Lost City Hydrothermal Field**
   Scope Definition: The actively venting carbonate-brucite chimneys and the immediately issuing hydrothermal fluids of the Lost City field (excluding the deep sub-seafloor basement or the surrounding unaltered open-ocean seawater).

   a. Temperature (°C): 40 to 116
      Source: Group A says the venting fluids reach a maximum of 90–91°C because early sampling missions and temperature probes recorded this as the peak sustained temperature at the orifices (Kelley et al., 2001, Proskurowski et al., 2006). Group B says the temperatures can reach up to 116°C because later, more advanced isobaric gas-tight samplers deployed specifically at the Beehive structure successfully captured hotter, pristine fluids directly from the vent interior (Seyfried et al., 2015). A union range of 40–116°C is adopted because it accounts for both the diffusely venting cooler fluids and the hottest focused flow.

      Reliability: 3 (High). The data are derived from direct field measurements utilizing calibrated sensors and isobaric gas-tight samplers deployed by remotely operated vehicles.

   b. pH: 9 to 11
      Source: In-situ measurements and geochemical models establish that the vent fluids are highly alkaline, consistently ranging from pH 9 to 11 (Kelley et al., 2001). Other sources land on slightly tighter ranges, such as 9 to 10.8, depending on the degree of seawater mixing and the specific chimney sampled (Ludwig et al., 2006). Disagreement is minimal and purely a function of whether pH is measured directly in-situ or extrapolated to standard shipboard conditions (Seyfried et al., 2015).

      Reliability: 3 (High). The pH values were obtained from primary mission data using direct field measurements published in peer-reviewed journals.

c. Salinity (g/kg): ~35
Source: Analysis shows that the hydrothermal fluids exhibit a salinity nearly identical to ambient seawater, with chloride concentrations around 540 to 554 mmol/kg (Seyfried et al., 2015). Some earlier sources land on slightly depleted chloride levels due to minor hydration effects or chloride uptake by serpentine minerals (Allen & Seyfried, 2004). The nature of this minor disagreement stems from the challenges of separating trace seawater entrainment from the true hydrothermal endmember, but all agree salinity fundamentally mirrors seawater.

Reliability: 3 (High). Major ion concentrations are verified through direct mass spectrometry of recovered fluid samples in peer-reviewed literature.

d. Pressure (MPa): 7.5 to 9.0
Source: Peer-reviewed sources establish that the field sits at a bathymetric depth of 750 to 900 meters, which equates directly to a hydrostatic pressure of approximately 7.5 to 9.0 MPa (Kelley et al., 2001). Other researchers studying subseafloor circulation use pressures up to 15 MPa to model the deeper roots of the system (Seyfried et al., 2015). The apparent disagreement merely reflects the difference between the seafloor vent depth and the deeper sub-surface reaction zone.

Reliability: 3 (High). Pressure is a direct physical conversion from the well-documented bathymetric depth of the hydrothermal field.

e. Isolation Score (1-5): 1
(i) Physical Confinement (1-5) 1
Source: The chimneys vent directly into the open ocean, resulting in immediate physical mixing with the surrounding water mass (Kelley et al., 2005). The system is entirely open water without any ice or rock barriers enclosing the venting fluid.

(ii) Hydrologic Connectivity (1-5): 1
Source: Venting fluids freely exchange with the deep Atlantic Ocean waters, creating broad geochemical plumes (Kelley et al., 2001). This open boundary ensures the habitat is continuously connected with the global oceanic water mass with free exchange.

(iii) Isolation Time (1-5): 1
Source: Once the fluids exit the chimney structures, they undergo instantaneous, open exchange with ambient seawater (Ludwig et al., 2006). The isolation time for the vent habitat surface is effectively zero years due to this constant, uninhibited free exchange.

Reliability: 3 (High). The open-ocean setting and fluid exchange dynamics are extensively verified by direct submersible observations in peer-reviewed journals.

f. Redox Score (1 -5): 4.67
(i) Redox Species (1-5): 4
Source: The fluids contain exceptionally high concentrations of reduced species like H_2 and CH_4, alongside abundant seawater-derived sulfate, but lack significant dissolved CO_2 (Proskurowski et al., 2008). This provides strong evidence for both highly potent oxidants and reductants, making it an extremely rich chemical environment.

(ii) Equilibrium (1-5): 5
Source: Mixing between highly reducing, high-pH vent fluids and oxygenated seawater creates a well-documented constant flux and steep, exploitable redox gradient (Kelley et al., 2005). The system's redox potential is estimated to be incredibly stable around -800 mV, driving continuous disequilibrium (Schwander et al., 2023).

(iii) Chemolithotrophic Metabolism (1-5): 5
Source: There is strong, confirmed evidence of dense biofilms dominated by chemolithotrophic *Methanosarcinales* and sulfate-reducing bacteria thriving on the available gradients (Schrenk et al., 2004). This biology is directly and dominantly fueled by the geochemical disequilibrium generated by the venting fluids.

Reliability: 3 (High). The redox conditions and microbial metabolisms are thoroughly confirmed by direct sampling, genomic sequencing, and isotopic analysis.

3. **Strýtan Hydrothermal Field**
   Scope Definition: The site encompasses the shallow submarine alkaline hydrothermal vent chimneys, primarily Big Strýtan and Arnarnesstrýtan, located at depths of 15 to 60 meters in the Eyjafjördur fjord in northern Iceland, including their actively discharging meteoric-derived fluids (Stanulla et al., 2017, Twing et al., 2022).

   a. Temperature (°C): 66 to 78.1
      Source: The most comprehensive geochemical surveys establish that the maximum hydrothermal fluid temperatures at the Strýtan Hydrothermal Field reach approximately 70 °C (Twing et al., 2022) or even up to 78.1 °C (Price et al., 2017). Other regional geothermal surveys note that the broader Eyjafjördur region has waters reaching up to 90 °C (Kristmannsdóttir & Johnsen, 1982). There is no significant disagreement in the literature regarding these in-situ temperature profile measurements for the primary active venting chimneys.

      Reliability: 3 (High) Temperature data originates from direct field measurements utilizing calibrated sensors during peer-reviewed scientific diving expeditions.

b. pH: 8.1 to 10.2
Source: In-situ measurements confirm that the hydrothermal fluids at Strýtan have pH values ranging from 8.1 to 10.2 (Price et al., 2017), with some suggesting they are more highly alkaline, with pH values ranging from 9.4 at Arnarnesstrýtan to 10.2 at Big Strýtan (Twing et al., 2022). Other studies corroborate this hyperalkaline nature, universally identifying the site as an alkaline submarine hydrothermal system (Stanulla et al., 2017). The minor variance in the reported pH range represents natural differences between specific individual chimney structures across the vent field rather than any methodological disagreement among scientists.

Reliability: 3 (High). Direct in-situ pH measurements were systematically recorded during peer-reviewed field sampling campaigns utilizing precise geochemical analytical methods.

c. Salinity (g/kg): ~0.17 to 5.07
Source: Group A says the hydrothermal fluid is essentially fresh meteoric groundwater because geochemical analyses show exceptionally low chloride concentrations of 1.3 to 21.6 mM compared to the 530 mM of the surrounding subarctic seawater (Twing et al., 2022). Group B says the measured salinity can appear slightly higher because the freshwater undergoes varying degrees of mixing with oceanic water within the highly permeable chimney structures before emission (Stanulla et al., 2017). The union range adopted reflects this pristine freshwater origin, acknowledging that any minor salinity fluctuations purely reflect the degree of seawater entrainment occurring immediately prior to the sampling process rather than the source fluid composition (Kristmannsdóttir & Johnsen, 1982).

Reliability: 3 (High). Salinity data relies on peer-reviewed geochemical analyses of major ions from directly sampled venting.

d. Pressure (MPa): 0.25 to 0.7
Source: The best available primary sources report that the Strýtan hydrothermal chimneys are located in shallow coastal waters at depths ranging from 15 to 60 meters below level (Stanulla et al., 2017). Other investigations independently corroborate this specific depth profile when describing the giant submarine smectite cones in the Eyjafjördur fjord (Marteinsson et al., 2001). There is no fundamental disagreement regarding the pressure environment, as the minor depth discrepancies purely represent measurements taken from different chimney structures within the same shallow marine ecosystem.

Reliability: 3 (High). Pressure is derived directly from well-documented depth measurements confirmed by scientific scuba divers on site.

e. Isolation score (1-5): 1
(i) Physical Confinement (1-5): 1
Source: The Strýtan hydrothermal chimneys are situated in the open, shallow marine waters of the Eyjafjördur fjord (Twing et al., 2022). This constitutes an open water environment characterized by free exchange with the surrounding subarctic seawater (Stanulla et al., 2017).

(ii) Hydrologic Connectivity (1 -5): 1
Source: The hydrothermal system is fed by meteoric groundwater that discharges directly into the ocean (Kristmannsdóttir & Johnsen, 1982). This represents a high degree of hydrologic connectivity with free mass exchange into a large marine body (Marteinsson et al., 2001).

(iii) Isolation Time (years) (1 -5): 1
Source: The hydrothermal fluid exhibits a continuous flow into the ocean, allowing for immediate dispersal (Stanulla et al., 2017). Consequently, there is an open exchange timescale of essentially zero years upon discharge into the fjord (Twing et al., 2022).

Reliability: 3 (High) Based on direct geological observations and peer-reviewed mapping of the vent field's open marine setting.

f. Redox Score (1 -5): 4
(i) Redox Species (1 -5): 4
Source: Primary sources report strong evidence for diverse redox species, notably the mixing of hydrothermal hydrogen sulfide with oxygenated, sulfate-rich seawater (Twing et al., 2022). Other studies corroborate these steep chemical gradients generated by the alkaline discharges (Stanulla et al., 2017).

(ii) Equilibrium (1 -5): 4
Source: Investigations demonstrate that mixing alkaline, fresh hydrothermal fluids with cold seawater creates steep pH and sodium gradients (Twing et al., 2022). Researchers emphasize that these sustained chemical disequilibria represent a strong thermodynamic gradient for life (Marteinsson et al., 2001).

(iii) Chemolithotrophic Metabolism (1 -5): 5
Source: Peer-reviewed sequencing confirms the presence of chemolithoautotrophs and sulfur-oxidizing bacteria, including taxa utilizing the Wood-Ljungdahl pathway (Twing et al., 2022). Thermodynamic modeling additionally proves that catabolic reactions are highly exergonic and support biological activity here (Rucker et al., 2023).

Reliability: 3 (High) because the redox assessments are supported by in-situ geochemical sampling and genetic amplicon sequencing published in peer-reviewed journals.

4. **Lake Vostok, Antarctica**
   Scope Definition: This site includes the main freshwater body of Lake Vostok, the shallow embayment, and the associated subglacial sediments and accreted ice interface beneath the East Antarctic Ice Sheet, excluding the overlying meteoric ice.

   a. Temperature (°C): -2.53 to -3.07
      Source: The best limnological and modeling sources indicate the surface waters of Lake Vostok are at the pressure melting point, ranging from -2.53 C to -3.07 C depending on ice thickness (Wüest & Carmack, 2000; Thoma et al., 2008). Other remote sensing estimates and extrapolations agree with this thermodynamically constrained range, demonstrating high consensus across disciplines (Christner et al., 2006). There is effectively no disagreement regarding this parameter, as it is governed by established physical laws of water under immense pressure beneath the ice sheet (Siegert et al., 2001).

      Reliability: 3 (High). The score is based on peer-reviewed physical models and thermodynamic calculations directly tied to highly precise geophysical measurements of the overlying ice thickness.

   b. pH: ~6.0
      Source: Geochemical models based on accretion ice samples suggest the surface lake waters have a slightly acidic pH of approximately 6.0 (Christner et al., 2006). Other extrapolated data from overlying meteoric ice confirm that subglacial Antarctic lakes generally maintain a slightly acidic character due to the lack of atmospheric buffering (Siegert et al., 2012). There is minimal disagreement, although researchers note that exact values may vary locally depending on potential hydrothermal input or weathering reactions at the lakebed (Siegert et al., 2003).

      Reliability: 3 (High). This score relies entirely on peer-reviewed geochemical modelling derived from primary ice core sample data and established chemical principles.

   c. Salinity (g/kg): 0.0015 to 1.2
      Source: Group A says the lake is extremely fresh, between 0.0015 and 0.034 g/kg, because total dissolved solids estimated from accretion ice partitioning are very low (Christner et al., 2006). Group B says salinity could be higher, reaching 0.4 to 1.2 g/kg, because significant unfractionated lake water might be entrained during freezing, causing previous partitioning models to underestimate salt content (Siegert et al., 2003). The union range of 0.0015 to 1.2 g/kg is adopted because both groups base their models on the identical Vostok accretion ice data but apply different, equally plausible physical assumptions about solute rejection during the freezing process (Siegert et al., 2001).

      Reliability: 3 (High). This score reflects that all measurements stem from peer-reviewed analyses of primary accretion ice cores, despite varying interpretative models.

d. Pressure (MPa): ~35.5
Source: Glaciological and geophysical surveys calculate the lake's pressure to be roughly 35.5 MPa, driven by the overburden of 3,750 to 4,150 meters of glacial ice (Siegert et al., 2001). Other independent modeling studies completely corroborate this, citing a functionally identical pressure regime governed strictly by ice thickness and density (Wüest & Carmack, 2000). There is no disagreement regarding this parameter, as it is a direct hydrostatic calculation from extensively validated airborne radar and satellite altimetry measurements (Kapitsa et al., 1996).

Reliability: 3 (High). This high score is supported by extensively peer-reviewed, highly accurate remote sensing data and fundamental hydrostatic physics.

e. Isolation Score (1-5): 5
(i) Physical Confinement (1-5): 5
Source: The lake is physically confined beneath nearly four kilometers of solid glacial ice and sits within a deep continental bedrock basin, creating a highly sealed, deep subsurface environment completely cut off from the atmosphere (Christner et al., 2006; Siegert et al., 2001).

(ii) Hydrologic Connectivity (1-5): 5
Source: Lake Vostok is located in a closed subglacial topographic basin with no known outflow, and this extreme hydrologic isolation prevents any free exchange of liquid water with outside systems (Siegert et al., 2001; Christner et al., 2006).

(iii) Isolation Time (1-5): 5
Source: Peer-reviewed estimates confirm that Lake Vostok has been physically isolated from the Earth's atmosphere for more than fifteen million years, a vast timescale that comfortably exceeds the one million year threshold required for the maximum isolation rating (Christner et al., 2006; Siegert et al., 2001).

Reliability: 3 (High). This score is justified by robust, peer-reviewed geological and glaciological evidence documenting the ice sheet's long-term stability and the lake's closed topography.

f. Redox Score (1-5): 4
(i) Redox Species (1-5): 4
Source: Melting glacial ice continuously delivers abundant oxygen via dissolving air clathrates while deep sediments likely host reduced species like metal sulfides, providing strong evidence for both oxidized and reduced chemical species (Siegert et al., 2001; Christner et al., 2006).

(ii) Equilibrium (1-5): 4
Source: The spatial separation between continuous oxygen input from northern melting and sulfide inputs from deep basal sediments creates a strong, exploitable geochemical gradient that guarantees the system remains out of thermodynamic equilibrium (Christner et al., 2006; Siegert et al., 2003).

(iii) Chemolithotrophic Metabolism (1-5): 4
Source: Viable bacteria identified in the accretion ice possess DNA signatures linking them to chemolithoautotrophic metabolisms capable of iron and sulfur oxidation, providing possible evidence that such microbial pathways are actively exploited (Bulat et al., 2004; Siegert et al., 2003).

Reliability: 3 (High). This score relies on peer-reviewed microbiological and geochemical analyses of accretion ice, which serve as direct, reliable proxies for the lake's internal redox conditions.

5. **Lake Untersee, Antarctica**
   Scope Definition: This site is limited to the deep, chemically stratified, anoxic southern basin (trough) of Lake Untersee at depths between 70 and 100 metres, explicitly excluding the well-mixed, oxic northern basin. This specific sub-environment is targeted because its physical isolation, permanent ice cover, and active hydrogenotrophic methanogenesis make it a highly relevant analogue.

   a. Temperature (°C): 3.7 to 5.0
      Source: Group A says temperatures reach a maximum of 5.0 °C at 70 metres depth because of solar heating through the clear water column (Wand et al., 1997), and drop to 3.7 °C near the 100-metre bottom because of the lack of convective mixing and differing density gradients (Bevington et al., 2018). The union range of 3.7 to 5.0 °C is adopted because it captures the full thermal profile of the anoxic basin below the chemocline.

      Reliability: 3 (High). The data originates from calibrated, in-situ multi-parameter sonde measurements published in peer-reviewed scientific journals.

   b. pH: 6.1 to 7.0
      Source: Limnological studies state that the anoxic bottom waters of the southern basin exhibit a slightly acidic to circumneutral pH ranging from 6.1 to 7.0 (Wand et al., 1997). Other sources focusing on the main oxic basin report a highly alkaline pH of 10.6, which does not apply to the isolated anoxic trough (Faucher et al., 2019). The nature of this disagreement is purely spatial, stemming from the lake's distinct sub-basin chemical stratification.

      Reliability: 3 (High). The data relies on direct in-situ water quality sonde measurements successfully published in rigorous peer-reviewed literature.

c. Salinity (g/kg): 0.57 to 0.88
Source: Hydrochemical analyses report Total Dissolved Solids ranging from 0.57 to 0.88 g/kg in the deepest parts of the anoxic zone (Wand et al., 1997). Other regional studies generally land on similar measurements, noting a sharp increase in salinity compared to the overlying oxic waters (Bevington et al., 2018). The nature of any disagreement is negligible, as the observed specific conductivity gradients consistently support this solute concentration without conflicting data.

Reliability: 3 (High). The values are derived from direct ion-chromatography and in-situ conductivity measurements in peer-reviewed studies.

d. Pressure (MPa): ~1.1
Source: Bathymetric surveys state that the anoxic southern basin reaches a maximum depth of approximately 100 metres, equating to roughly 1.1 MPa of hydrostatic pressure (Wand et al., 1997). Later glaciological sources uniformly land on this established depth profile for their thermal and biogeochemical modeling (Wand et al., 2006). The nature of any disagreement is non-existent, as published literature is in complete consensus regarding the basin's maximum depth and resulting physical pressure.

Reliability: 3 Depth and pressure are based on direct bathymetric soundings and sensor deployments.

e. Isolation score (1-5): 3
(i) Physical Confinement (1-5): 3
Source: This physical barrier provides confinement by preventing wind-driven mixing and severely restricting light penetration to the benthic ecosystem beneath an ice cover averaging 3.0 metres in thickness (Wand et al., 1997, Andersen et al., 2011).

(ii) Hydrologic Connectivity (1 -5): 5
Source: The system is highly sealed due to the lack of surface outlets or a summer moat, creating a closed-basin environment completely cut off from atmospheric gas exchange and fed exclusively by subglacial meltwater (Faucher et al., 2019, Marsh et al., 2020).

(iii) Isolation Time (years) (1 -5): 2
Source: Stable water isotope modelling indicates the lake has remained well-sealed with no summer moating for the past 300 to 500 years, with isolation occasionally reset by periodic glacial lake outburst floods (Faucher et al., 2020, Faucher et al., 2021).
Reliability: 3 These metrics are supported by robust isotopic modelling and direct satellite observations in peer-reviewed journals.

f. Redox Score (1 -5): 5
(i) Redox Species (1 -5): 5
Source: Metagenomic analyses and biogeochemical sampling provide a clear physical basis for hydrogenotrophic methanogenesis by confirming the presence of high concentrations of dissolved methane alongside carbon dioxide and molecular hydrogen (Wand et al., 2006, Wagner et al., 2022).

(ii) Equilibrium (1 -5): 4
Source: The highly stratified water column physically traps reduced chemical species like methane and hydrogen sulfide at depth beneath an oxygenated epilimnion, creating a sharp, highly exploitable chemocline gradient (Bevington et al., 2018, Wagner et al., 2022).

(iii) Chemolithotrophic Metabolism (1 -5): 5
Source: The anaerobic microbial community utilizes specific chemolithotrophic pathways to survive extreme conditions, with advanced genomic and isotopic studies providing conclusive evidence of active hydrogenotrophic methanogenesis and sulfate reduction (Wand et al., 2006, Wagner et al., 2022).

Reliability: 3 The evaluations rely on metagenomic sequencing and radiotracer experiments published in top-tier journals.

6. **Lost Hammer Spring**
   Scope Definition: This site includes the perennial, hypersaline, subzero spring outlet—specifically the salt tufa dome and underlying vent sediments—on Axel Heiberg Island, Nunavut. It explicitly excludes the downstream runoff channels, adjacent tundra, and nearby but geochemically distinct analogue sites such as Colour Peak and Gypsum Hill springs.

   a. Temperature (°C): -5.9 to -4.7
      Source: The most rigorous multi-year, in-situ investigations confirm that the Lost Hammer spring outlet maintains a highly stable perennial temperature of approximately -5°C. Other earlier or broader studies sometimes report a slightly wider range of -5.9°C to -4.7°C depending on the exact season and specific micro-location within the vent. Any minor disagreement in these recorded temperatures generally arises from the difference between highly insulated core vent brines and peripheral surface fluids exposed to extreme seasonal air temperature fluctuations (Niederberger et al., 2010, Lay et al., 2013, Magnuson et al., 2022).

      Reliability: 3 (High). The reported temperatures are derived directly from calibrated field sensors deployed in the brine during multiple expeditions in both winter and summer seasons.

b. pH: 5.96 to 7.38
   Source: Researchers measuring pH at the Lost Hammer spring have reported a somewhat contested range between 5.96 and 7.38. Group A says slightly acidic values closer to 5.96 to 6.8 because their specific sampling events captured winter conditions where the spring is restricted, concentrating dissolved constituents and dissolved gases. Group B says near-neutral to slightly alkaline values up to 7.38 because their summer sampling coincided with partial atmospheric exposure, gas exsolution, and varying flow dynamics. A union range is adopted because it accurately reflects the genuine seasonal and spatial variability of the spring's dynamic surface geochemistry across different field campaigns (Lay et al., 2012, Battler et al., 2013).

   Reliability: 3 (High). Measurements were taken in-situ using calibrated multi-probe systems directly in the spring water over several years.

c. Salinity (g/kg): 220 to 260
   Source: The consensus among leading microbiological and geochemical studies establishes the outlet's salinity at 24%, or approximately 240 g/kg, classifying it as a hypersaline brine. Alternative measurements report a tight range between 220 and 260 g/kg based on different seasonal evaporation and freezing concentration rates. Discrepancies are minimal and exclusively reflect the natural, seasonal evaporative and cryo-concentration cycles that fractionate the brine as it reaches the surface (Niederberger et al., 2010, Magnuson et al., 2022).

   Reliability: 3 (High). Salinity is consistently corroborated by primary analytical field data and rigorous laboratory ion chromatography.

d. Pressure (MPa): 0.1
   Source: Field reports confirm that the sampled spring outlet and tufa structures exist at the surface, subject to standard atmospheric pressure of approximately 0.1 MPa. Other sources detailing the hydrogeological origins note that the fluids ascend from deep beneath 400 to 600 metres of permafrost, where pressures are significantly higher. There is no fundamental disagreement regarding the surface pressure, though the subsurface source reservoir undeniably experiences elevated hydrostatic and lithostatic pressures prior to discharge (Pollard et al., 1999, Andersen et al., 2002).

   Reliability: 3 (High). This metric relies on the observable physical fact of a subaerial spring environment assessed by direct field measurement.

e. Isolation Score (1-5):
(i) Physical Confinement (1-5): 2
Source: The spring outlet features a partially confined salt tufa dome that overflows and is periodically blocked by freezing and precipitation during the harsh winter months, thereby significantly restricting open continuous flow (Battler et al., 2013).

(ii) Hydrologic Connectivity (1-5): 2
Source: The spring is fed by a restricted subpermafrost aquifer but discharges directly into a surface channel and ultimately connects to local river systems, allowing for seasonal exchange with surrounding bodies (Lay et al., 2013).

(iii) Isolation Time (1-5): 1 Source: Once the deep fluid emerges at the surface spring outlet, it experiences continuous open atmospheric gas exchange and active seasonal surface flow, representing essentially zero isolation time at the surface (Niederberger et al., 2010).

Reliability: 2 (Medium). Assessed via general geomorphological observations and aerial surveys rather than precise subsurface hydrogeological tracer testing.

f. Redox Score (1-5):
(i) Redox Species (1-5): 5
Source: High-resolution gas and fluid chromatography definitively confirm the presence of strong oxidants like sulfate alongside copious reduced species, specifically including methane, hydrogen sulfide, and both molecular hydrogen and carbon dioxide (Lamarche-Gagnon et al., 2015).

(ii) Equilibrium (1-5): 4
Source: In-situ probes consistently record highly reducing, stable conditions with an oxidation-reduction potential around -165 mV to -187 mV, providing strong evidence for a highly exploitable chemical gradient within the system (Magnuson et al., 2022).

(iii) Chemolithotrophic Metabolism (1-5): 5
Source: Multi-omics and culture-based studies provide strong, confirmed evidence of a highly active chemolithoautotrophic microbial community that is biologically driven by active anaerobic methane oxidation, sulfide oxidation, and sulfate reduction processes (Magnuson et al., 2022).

Reliability: 3 Data is supported by direct molecular assays, RNA/DNA sequencing, calibrated in-situ multi-probe geochemical measurements and primary mission-equivalent data.

7. **Blood Falls, Antarctica**
   Scope Definition: The subglacial and englacial brine reservoir beneath Taylor Glacier, Antarctica, specifically encompassing the undisturbed hypersaline liquid prior to atmospheric alteration at the surface discharge point.

   a. Temperature (°C): -7.1 to -5.2
      Source: In situ englacial measurements record the subglacial brine temperature at -7.1°C, while the ambient glacier ice is approximately -17°C. Other surface-based discharge samples record slightly warmer temperatures between -5.6°C and -5.2°C. The nature of this minor disagreement stems from sampling location, as the brine undergoes latent heat transfer and mixes with varying thermal profiles as it flows through the englacial conduit towards the surface (Badgeley et al., 2017, Lyons et al., 2019, Mikucki et al., 2009).

      Reliability: 3 (High). Direct englacial probe measurements and repeated surface sampling firmly establish the persistent sub-zero liquid thermal state of the brine.

   b. pH: 6.2 to 9.3
      Source: Group A says the pH is circumneutral, around 6.2, because samples collected directly from uncontaminated outflows and englacial conduits reflect the suboxic, sealed subglacial reservoir. Group B says the pH is highly basic, ranging from 8.0 to 9.3, because measurements taken further down the outflow fan reflect rapid geochemical changes as the brine interacts with the atmosphere. A union range from 6.2 to 9.3 is adopted because the brine rapidly releases inorganic carbon and oxygenates upon atmospheric exposure, causing an immediate pH rise during surface transit (Campen et al., 2019, Mikucki et al., 2004, Mikucki et al., 2009, Sklute et al., 2022).

      Reliability: 2 (High). Extreme rapid geochemical alteration upon surface exposure complicates isolating the exact in situ subglacial pH without specialized englacial drilling.

   c. Salinity (g/kg): 100 to 126
      Source: The most direct englacial measurements indicate a total dissolved solids concentration of 126 g/L, identifying it as a hypersaline brine of marine origin. Other sources report slightly lower salinity ranges around 100 to 110 g/L or approximately 8% to 9.7% salinity for surface outflow samples. The nature of this disagreement arises from periodic dilution with fresh supraglacial meltwater during surface discharge events, whereas deep englacial samples remain undiluted (Campen et al., 2019, Lyons et al., 2005, Lyons et al., 2019).

      Reliability: 3 (High). Repeated multi-year chemical analyses consistently confirm hypersalinity, verifying the origin as cryoconcentrated ancient seawater.

d. Pressure (MPa): 2.6
Source: The best glaciological models estimate that the excess subglacial pressure driving the artesian brine release is approximately 2.6 MPa. Other sources focus on the atmospheric pressure at the discharge point, where the fluid emerges unpressurized at the glacier snout. The nature of this disagreement represents the spatial difference between the highly pressurized subglacial aquifer trapped beneath hundreds of meters of ice and the depressurized surface environment where the Blood Falls discharge occurs (Badgeley et al., 2017, Carr et al., 2022, Mikucki et al., 2004).

Reliability: 2 (Medium). The 2.6 MPa value is derived from hydraulic-potential modelling rather than direct in situ subglacial pressure transducer measurements.

e. Isolation Score (1-5): 4
(i) Physical Confinement (1-5): 4
Source: The isolated brine resides within a deep subglacial aquifer permanently trapped beneath Taylor Glacier and regional permafrost, physically requiring intermittent localized crevassing acting as a pressure-release valve for subaerial exposure (Badgeley et al., 2017).

(ii) Hydrologic Connectivity (1-5): 3
Source: While highly isolated from modern surface meltwater influences, the subglacial aquifer remarkably maintains an extensive, deep groundwater network that hydraulically connects directly to the deep bottom waters of Lake Bonney (Mikucki et al., 2015).

(iii) Isolation Time (years) (1-5): 5
Source: Geochemical and isotopic evidence firmly indicates the ancient marine brine has been completely isolated from the atmosphere for at least 50,000 years, and potentially up to 1.5 to 5 million years (Fisher & Schulze-Makuch, 2013, Mikucki et al., 2009).

Reliability: 3 Stable isotope data and the absence of phototrophic microorganisms robustly confirm profound, long-term isolation from the contemporary surface environment.

f. Redox Score (1-5): 5
(i) Redox Species (1-5): 5
Source: The enclosed subglacial brine is profoundly anoxic and suboxic, serving as a vast subsurface reservoir of reduced iron and sulfate, whilst completely lacking dissolved oxygen and nitrate for microbial use (Lyons et al., 2019, Mikucki et al., 2009).

(ii) Equilibrium (1-5): 4
Source: The deep subsurface environment maintains a delicate chemical equilibrium at an Eh of approximately 90 to 100 mV, which violently dis-equilibrates into rapid iron oxidation immediately upon surface atmospheric exposure (Campen et al., 2019, Mikucki et al., 2004).

(iii) Chemolithotrophic Metabolism (1-5): 5
Source: The subglacial ecosystem is fundamentally sustained by a unique cryptic sulfur cycle, where microbes thermodynamically exploit the subglacial bedrock via iron-mediated sulfate reduction in the complete absence of solar energy (Fisher & Schulze-Makuch, 2013, Mikucki et al., 2009).

Reliability: 3 Genomic sequencing and direct biogeochemical assays have conclusively demonstrated active chemolithoautotrophic pathways exploiting the iron and sulfur gradients (Mikucki et al., 2009).

8. **Lake Bonney, Antarctica (East Lobe)**
   Scope Definition: Perennially ice-covered, closed-basin meromictic lake located in the Taylor Valley, Antarctica, excluding the West Lobe and separated from it by a 12-meter deep sill. The scope encompasses the freshwater mixolimnion and the hypersaline, suboxic monimolimnion down to approximately 40 meters in depth.

   a. Temperature (°C): -2.6 to 7.5
      Source: Studies indicate a highly stratified thermal profile, with a maximum temperature around 6.3°C to 7.5°C occurring at 13 to 16 meters depth, and minimum temperatures of -2.0°C to -2.6°C in the deep bottom waters. Older studies sometimes recorded slightly different maxima, illustrating that temperature profiles have adjusted over time due to rising lake levels and heat transfer dynamics. Disagreements in exact historical peak temperatures are minor and primarily reflect the temporal evolution of the lake's thermal structure responding to changing water levels rather than measurement errors (Spigel & Priscu, 1996, Matsubaya et al., 1979, Hendy et al., 1977).

      Reliability: 3 (High). Decades of continuous depth-profile measurements utilizing calibrated thermistors and CTD casts demonstrate consistent structural thermal trends.

   b. pH: 5.5 to 7.0
      Source: Studies show that the fully oxygenated portions of the East Lobe maintain a fairly constant pH ranging between 6.0 and 7.0 across most depths. Other observations specifically pinpoint a slightly more acidic pH value of 5.5 occurring precisely at the 5-meter depth mark. There is virtually no scientific disagreement regarding these values, as the narrow pH range reflects the lake's relatively stable buffering capacity and lack of robust hydrogen sulfide-driven acidification found in other anoxic basins (Burton, 1981).

      Reliability: 3 (High). Measurements were taken directly from established vertical water column profiles, showing consensus across standard limnological methodologies.

c. Salinity (g/kg): 10 to 298.2
Source: Sources characterize the East Lobe as a strongly stratified meromictic lake, transitioning from fresh meltwater directly beneath the ice cover to extreme hypersalinity in the bottom waters. Group A reports absolute chloride ion concentrations reaching 196,000 ppm at the bottom, while Group B cites a total salinity of 298.2 g/kg at 32.5 meters depth. The union range adopted encompasses the freshwater mixolimnion (salinity <10 g/kg) down to the hypersaline monimolimnion (salinity >120 to ~298 g/kg), reflecting both the steep halocline below 20 meters and the complex evolutionary history of evaporative concentration and subsequent freshwater flooding (Hendy et al., 1977, Burton, 1981, Lee et al., 2004).

Reliability: 3 (High). Extensive, repeated geochemical profiling confirms extreme hypersalinity and strong chemical stratification.

d. Pressure (MPa): 0.1 to 0.5
Source: Topographic and bathymetric sources define the maximum depth of the East Lobe at approximately 36 to 40 meters beneath the 3 to 4.5 meter permanent ice cover. Applying standard hydrostatic pressure calculations to this depth profile yields a maximum pressure of approximately 0.4 to 0.5 MPa at the lake floor. There is no disagreement in the literature regarding the lake's bathymetry, though slight variations in maximum depth exist due to decades of gradually rising lake levels driven by glacial meltwater inputs (Spigel & Priscu, 1996, Lee et al., 2004, Hendy et al., 1977).

Reliability: 3 (High). Pressure is a direct physical derivation from well-documented bathymetric depths and known water densities.

e. Isolation score (1-5): 4
(i) Physical Confinement (1-5): 4
Source: The East Lobe is physically confined within a steep-sided, closed drainage basin and is capped year-round by a 3 to 4.5-meter thick permanent ice cover (Spigel & Priscu, 1996, Lee et al., 2004).

(ii) Hydrologic Connectivity (1-5): 3
Source: While the deep monimolimnion is strictly isolated, the upper water column receives intermittent summer meltwater from local alpine glaciers and surface overflow from the West Lobe via a shallow 12-meter sill (Spigel & Priscu, 1996, Hendy et al., 1977).

(iii) Isolation Time (1-5): 4
Source: Radiocarbon dating of dissolved inorganic carbon in the deep waters indicates apparent ages up to 10,400 14C years BP, highlighting profound long-term isolation from the atmosphere (Doran et al., 2014).

Reliability: 3 (High). Physical constraints are visually and topographically confirmed, and isolation times are backed by robust radiocarbon and noble gas geochemical techniques.

f. Redox Score (1-5): 3
(i) Redox Species (1-5): 3
Source: The water column transitions from oxygen super-saturation in the upper layers to suboxic conditions in the hypersaline deep waters, lacking the complete anoxia or hydrogen sulfide found elsewhere (Torii et al., 1975, Lee et al., 2004).

(ii) Equilibrium (1-5): 4
Source: Strong vertical density stratification suppresses physical mixing, sustaining sharp, persistent disequilibrium gradients of dissolved gases, nutrients, and electron acceptors across the established chemocline (Spigel & Priscu, 1996, Priscu, 1997).

(iii) Chemolithotrophic Metabolism (1-5): 3
Source: Microbial ecosystems are restricted; denitrifiers are present but denitrification is thermodynamically unfavorable, leading to nitrous oxide buildup via nitrification in the suboxic deep layer (Priscu, 1997, Lee et al., 2004).

Reliability: 3 (High). Consistent detection of stable redox potentials and corresponds to microbial metabolic limitations across multiple independent geochemical surveys.

9. **Lake Bonney, Antarctica (West Lobe)**
   Scope Definition: Encompasses the permanently ice-covered, stratified water column of the West Lobe of Lake Bonney, from the ice-water interface down to the benthic floor (approximately 40 m depth), including the hypersaline, suboxic hypolimnion, but specifically excluding the East Lobe, the overlying Taylor Glacier, and the subglacial reservoir itself.

   a. Temperature (°C): -5.35 to 3.1
      Source: Group A (Angino, Armitage & Tash, 1964; Hendy et al., 1977) says temperature ranges from -5.35 °C at the bottom to a maximum of 1.35 to 2.2 °C just below the ice because of historical solar heating and shallower depth constraints in the 1960s. Group B (Spigel & Priscu, 1996) says the maximum temperature reaches 3.1 °C with a minimum of -5.0 °C because rising lake levels have thickened the freshwater lens, altering solar radiation absorption and heat transfer since the earlier measurements. A union range of -5.35 to 3.1 °C is adopted because it accurately encompasses both historical baseline and contemporary thermal dynamics within the isolated lake system.

      Reliability: 3 (High). Multiple in-situ conductivity-temperature-depth casts over decades demonstrate consistent thermal stratification with minor variations attributed to measurable lake level rise.

b. pH: 6.0 to 8.0
Source: Sources identify a distinct vertical gradient where the surface water pH is relatively neutral to slightly basic, around 7.5 to 8.0. Other observations from the deep hypersaline waters below the chemocline land at distinctly more acidic values, reaching a pH of approximately 6.0 near the bottom. There is no fundamental disagreement among authors, as the variance purely reflects the extreme biogeochemical stratification between the oxygenated upper layer and the suboxic hypolimnion. (Lee et al., 2004; Mikucki et al., 2004).

Reliability: 3 (High). Direct potentiometric measurements and consistent documentation of the acidic deep-water anomaly in independent peer-reviewed geochemical surveys.

c. Salinity (g/kg): 1 to 150
Source: Sources agree that the lake exhibits extreme stratification, transitioning from nearly fresh water above the 15-meter chemocline to hypersaline brine below with bottom salinities reaching 15%. Other measurements land at similar magnitudes, reporting deep chloride concentrations approaching 100 g/kg and total dissolved solids around 150 g/kg. Minor disagreements regarding exact salinity percentages reflect different sampling years, slightly different depth intervals, or variations in the specific equations of state used rather than a fundamental environmental shift. (Hendy et al., 1977; Matsubaya et al., 1979; Spigel & Priscu, 1996; Lee et al., 2004).

Reliability: 3 (High). Historical and contemporary hydrographic profiles repeatedly confirm the extreme, stable halocline using calibrated standard and modified state equations.

d. Pressure (MPa): 0.1 to 0.5
Source: Sources agree that the lake operates under standard atmospheric pressure at the surface beneath a 3.0 to 4.5-meter perennial ice cover. Other hydrographic models land at identical hydrostatic assumptions, calculating maximum pressures of approximately 0.5 MPa corresponding to the uniform 40-meter maximum basin depth. There is absolutely no disagreement concerning this parameter, as pressure gradients are dictated entirely by well-established bathymetry and the highly stable depth profile. (Spigel & Priscu, 1996; Lee et al., 2004; Obryk et al., 2016).

Reliability: 3 (High). Hydrostatic pressure is a direct physical derivation from extensively mapped, high-resolution bathymetric data and conductivity-temperature-depth profiles.

e. Isolation Score (1-5): 4
(i) Physical Confinement (1-5): 4
Source: The West Lobe is physically confined by steep valley walls, a bordering glacier terminus, and a 3-to-4.5-meter-thick perennial ice cover that drastically limits atmospheric exchange and sunlight. (Spigel & Priscu, 1996; Lee et al., 2004; Obryk et al., 2016).

(ii) Hydrologic Connectivity (1-5): 3
Source: Despite surface isolation, the hypolimnion maintains hydrologic connectivity via episodic inflows of hypersaline brine from a subglacial reservoir and eastward advection over a shallow 12-meter sill into the adjacent lobe. (Spigel & Priscu, 1996; Mikucki et al., 2004; Lawrence et al., 2020).

(iii) Isolation Time (1-5): 4
Source: Radiocarbon dating of dissolved inorganic carbon in the deep waters indicates an apparent isolation age of roughly 27,000 years before present, marking it as one of the most radiocarbon-deficient lacustrine environments identified. (Doran et al., 2014).
Reliability: 3 Comprehensive isotopic dating, subglacial hydraulic modeling, and long-term limnological monitoring provide consistent, independent evidence of confinement.

f. Redox Score (1-5): 4
(i) Redox Species (1-5): 4
Source: The aphotic deep waters are highly enriched in reduced iron, sulfate, and dimethylated sulfur compounds, establishing a strictly suboxic to anoxic environment beneath the oxycline. (Lee et al., 2004; Mikucki et al., 2004; Campen et al., 2019).

(ii) Equilibrium (1-5): 5
Source: The lake exhibits extreme, long-term density stratification driven by a steep halocline that prevents vertical mixing and locks extreme biogeochemical disequilibrium between the photic surface and aphotic bottom. (Spigel & Priscu, 1998; Lee et al., 2004; Lawrence et al., 2020).

(iii) Chemolithotrophic Metabolism (1-5): 3
Source: The suboxic deep layer supports slow chemolithotrophic processing, notably microbial sulfur cycling and iron oxidation, which utilize the subglacial brine inputs as vital metabolic substrates. (Lee et al., 2004; Mikucki et al., 2004).

Reliability: 3 (High). These specific redox conditions and microbial activities are repeatedly verified by rigorous enzymatic assays and high-resolution chemical profiling.

10. Ace Lake, Antarctica

Scope Definition: The perennially anoxic, sulfidic monimolimnion and associated benthic zones of Ace Lake, Antarctica, specifically isolated below the chemocline at depths of 11 to 25 metres, excluding the oxygenated upper mixolimnion.

a. Temperature (°C): 1 to 11.4
   Source: Limnological surveys state that the bottom waters of the permanently anoxic monimolimnion remain constantly between 1 °C and 2 °C year-round (Franzmann et al., 1991). Other physical profiles note that the lake features a mid-water thermal maximum that reaches up to 11.4 °C just above the chemocline due to solar heating, before steadily cooling towards the sediments (Gibson, 1999). There is no fundamental disagreement in the literature, merely a distinction between the seasonally variable, heliothermal upper layers and the highly stable, perennially cold conditions of the deep bottom waters (Rankin et al., 1999).

   Reliability: 3 (High). The data originate from peer-reviewed journals and direct field measurements using calibrated submersible data loggers and thermistors.

b. pH: 6.8 to 8.3
   Source: Geochemical studies establish that the pH in the oxic mixolimnion of the lake varies between 8.18 and 8.33 during the summer months (Rankin et al., 1999). Secondary observations confirm that beneath the chemocline, the anoxic monimolimnion exhibits a decreasing pH gradient that reaches 6.8 at the sediment interface (Burton, 1980). The sources universally agree on this parameter, with any minor variations attributed to the seasonal onset of primary productivity in the upper waters and the steady accumulation of acidic by-products in the stagnant bottom layer (Gibson et al., 1997).

   Reliability: 3 (High). pH values were determined through direct, calibrated field measurements and published in peer-reviewed scientific literature.

c. Salinity (g/kg): 7 to 43
   Source: The most authoritative limnological profiles demonstrate that the salinity of the monimolimnion increases sharply below the chemocline to reach a maximum of 43 g/kg near the bottom (Gibson, 1999). Other historical measurements align closely with these figures, citing an overall lake salinity range from 7 g/kg at the surface to 40 g/kg in the deep waters (Rankin et al., 1999). There is no disagreement regarding these concentrations; any minor differences reflect inter-annual variations in surface ice-melt or the precise depth of sampling rather than contradictory instrumentation or scientific methodologies (Burton, 1980).

   Reliability: 3 (High)This parameter is derived from peer-reviewed studies utilising direct, well-calibrated in-situ conductivity sensors and chemical titrations.

d. Pressure (MPa): 0.1 - 0.35
Source: The literature defines the maximum depth of the lake basin as 25 metres, which corresponds to approximately 0.35 MPa of absolute hydrostatic pressure at the sediment interface (Rankin et al., 1999). Additional bathymetric surveys consistently map the lake's deepest point between 23 and 25 metres depending on minor water level fluctuations (Gibson, 1999). There is no disagreement regarding this physical parameter, as the pressure is a direct, undisputed mathematical derivation from the maximum bathymetric depth firmly established across all long-term topographic studies of the region (Dartnall, 2000).

Reliability: 3 (High). Depth was measured using direct bathymetric surveys published in rigorous, peer-reviewed geographical and limnological literature.

e. Isolation Score (1-5):
(i) Physical Confinement (1-5): 2
Source: Rankin et al. (1999) confirm that the lake is covered by ice up to two metres thick for ten to twelve months annually. This warrants a score of 2 as it is partially confined by seasonal ice cover that occasionally melts out completely during warmer summers.

(ii) Hydrologic Connectivity (1-5): 3
Source: The lake lacks any outflow or groundwater flow, with water input strictly limited to highly restricted summer meltstreams and direct precipitation (Rankin et al., 1999). A score of 3 is appropriate due to this highly restricted exchange with external hydrological systems.

(iii) Isolation Time (years) (1-5): 3
Source: Palaeolimnological reconstructions indicate that the basin was completely isolated from the ocean by isostatic rebound approximately 3800 to 5000 years ago (Fulford-Smith & Sikes, 1996). A score of 3 fits perfectly since the isolation time falls directly within the 1000 to 100,000-year bracket.

Reliability: 3 Isolation metrics are heavily supported by precise, peer-reviewed radiocarbon dating, stable isotope analyses, and direct physical observations.

f. Redox Score (1-5):
(i) Redox Species (1 -5): 4
Source: The deep anoxic waters contain high concentrations of methane and hydrogen sulfide, alongside oxidants like sulfate in the upper layers (Franzmann et al., 1991). A score of 4 is given as there is strong evidence for an exploitable gradient of both oxidised and reduced chemical species.

(ii) Equilibrium (1-5): 4
Source: The redox potential drops sharply across the chemocline, reaching -167 mV in the monimolimnion due to hydrogen sulfide accumulation (Franzmann et al., 1991).

(iii) Chemolithotrophic Metabolism (1-5): 5
Source: Genomic analyses and pure microbial cultures have isolated active chemolithotrophic methanogens, such as Methanogenium frigidum, from the anoxic bottom waters (Wilkins et al., 2013). A score of 5 is applied because there is strong, confirmed evidence of organisms actively exploiting these chemical gradients for metabolism.

Reliability: 3 (High). Biogeochemical and microbiological findings derive from peer-reviewed genomic sequencing, targeted culturing, and in-situ chemical profiling.

11. Lake Enigma, Antarctica

Scope Definition: The subglacial liquid water column and benthic microbial mats located beneath the 9 to 14-metre perennial ice cover of Lake Enigma in the Northern Foothills of Victoria Land, Antarctica, excluding the surface ice and surrounding terrestrial environment.

a. Temperature (C°): -0.5 to 0.6
   Source: The most rigorous peer-reviewed sources report that the water column maintains a near-zero temperature profile, ranging from approximately -0.5 to 0.6 degrees Celsius (Smedile et al., 2024; Mercogliano et al., 2024). Grey literature and secondary science journalism pieces generally align with these primary measurements, discussing the permanently freezing ambient conditions without providing contradictory internal water temperatures (Hanks, 2025). There is no significant disagreement among the available sources, as all confirm the environment is a strictly cold-adapted, near-freezing liquid water body beneath the ice.

   Reliability: 3 (High). The temperature data is derived from direct, in-situ field measurements using calibrated sensors deployed through boreholes by peer-reviewed studies.

b. pH: 8.2 to 10.2
   Source: Limnological profiling of the lake demonstrates strong alkalinity, with pH values highly stratified from 10.0 to 10.2 in the upper layers, dropping to 8.2 to 8.8 near the lake bottom (Smedile et al., 2024). Other supporting microbiological studies corroborate this overarching condition, simply noting the lake's strong alkalinity without specifying the exact gradient depth profiles (Mercogliano et al., 2024). No sources contradict these findings, showcasing a uniform agreement on the pronounced alkaline nature of the entire water column.

   Reliability: 3 (High). This score reflects the use of direct, in-situ multiprobe sensor measurements published in high-impact, peer-reviewed journals.

c. Salinity (g/kg): 1.05 to 1.50
Source: Peer-reviewed analyses indicate that the lake is oligotrophic to slightly brackish, with total salinity ranging from 1.05 to 1.12 grams per litre in the metalimnion and up to 1.50 grams per litre in the deepest hypolimnion (Smedile et al., 2024). Secondary science reporting also confirms the presence of an isolated fresh to slightly brackish water layer, though they rarely provide exact quantitative salinity measurements (Sadigova, 2024). There is no disagreement among the sources, which uniformly characterise the lake as a low-salinity environment dominated by sodium and chloride ions.

Reliability: 3 (High). The salinity parameters are based on direct in-situ multiprobe measurements and subsequent rigorous laboratory ion chromatography from peer-reviewed literature.

d. Pressure (MPa): ~0.1 to 0.3
Source: Peer-reviewed geological surveys describe the lake as being sealed beneath 9 to 14 metres of solid ice, with the total lake depth reaching up to 23 metres, implying a hydrostatic pressure environment roughly between 0.1 and 0.3 megapascals (Smedile et al., 2024). Other literature focuses on the pressurised nature of the subglacial water inflow from the nearby glacier rather than providing exact hydrostatic pressure measurements for the lake body itself (Hanks, 2025). There is no disagreement, though direct pressure readings in megapascals are omitted in favour of depth measurements.

Reliability: 2 (High). While the depth measurements are highly accurate from peer-reviewed sources, the exact pressure in megapascals must be inferred mathematically rather than directly quoted.

e. Isolation Score (1-5): 4
(i) Physical Confinement (1-5): 4
Source: Peer-reviewed drilling records reveal a perennial ice shield between 9 and 14 metres thick that completely and tightly seals the liquid water column from the surface atmosphere (Smedile et al., 2024).

(ii) Hydrologic Connectivity (1-5): 4
Source: Isotopic and ground-penetrating radar data suggest the lake is fed by a restricted, hidden, and tightly closed subsurface drainage system originating from the nearby Amorphous Glacier (Smedile et al., 2024).

(iii) Isolation Time (years) (1-5): 3
Source: Biological evidence indicates the ecosystem consists of legacy biota trapped since the lake's ancient freeze-up, suggesting an extended isolation time exceeding seasonal exchanges, though an exact chronological age remains undetermined (Smedile et al., 2024; Paul, 2024).

Reliability: 3 (High). These physical and hydrological constraints are derived from high-resolution ground-penetrating radar and direct isotopic water sampling in peer-reviewed studies.

f. Redox Score (1-5): 4
(i) Redox Species (1-5): 4
Source: Genomic and in-situ sensor data confirm extreme supersaturation of dissolved oxygen alongside active nitrogen cycling species, indicating the presence of both strong oxidants and reduced nitrogen compounds (Smedile et al., 2024).

(ii) Equilibrium (1-5): 4
Source: The highly stratified water column maintains a sharp gradient from an oxygen-supersaturated metalimnion to a slightly hypoxic hypolimnion, providing a strongly exploitable thermodynamic gradient for microorganisms (Smedile et al., 2024).

(iii) Chemolithotrophic Metabolism (1-5): 5
Source: Peer-reviewed phylogenetic profiling provides strong, direct evidence of chemolithotrophy, specifically identifying nitrite-oxidising Nitrospira species and nitrite-reducing Gemmatimonadaceae actively participating in the lake's nitrogen cycle (Smedile et al., 2024).

Reliability: 3 (High). The redox and metabolic data are derived from state-of-the-art 16S rRNA amplicon sequencing and calibrated multiprobe sensors published in premier scientific journals.

12. Aurora Hydrothermal Vent Field

Scope Definition: This site encompasses the active hydrothermal vent system situated at roughly 3880 to 4000 metres depth on the ultraslow-spreading Gakkel Ridge in the Arctic Ocean. The scope includes the actively venting high-temperature black smokers (e.g., Hans Tore, Enceladus, Ganymede), surrounding areas of low-temperature diffuse fluid flow, inactive massive sulphide chimneys, and the overlying dispersing hydrothermal plume, all located beneath a perennial sea-ice cover (Ramirez-Llodra et al., 2023).

a. Temperature (°C): 50°C to 370
   Source: Primary mission data establishes that the Aurora vent field features high-temperature black smokers venting fluids between 350°C and 370°C, alongside diffuse flows measured between ~50°C and 126°C (German et al., 2022, Jakuba et al., 2024, Lapointe et al., 2025). Other general descriptions agree with these direct measurements, identifying distinct high-temperature clusters accompanied by extensive peripheral zones of lower-temperature discharge. There is no significant disagreement in the literature regarding these directly measured thermal gradients across the active field (Argentino et al., 2025, Walter et al., 2025).

   Reliability: 3 (High). Temperatures are derived from direct in-situ field measurements using calibrated sensors and published in peer-reviewed journals.

b. pH: ~5.50 to 6.06
Source: Expedition reports confirm that while precise pH figures for Aurora remain unpublished, geochemical analyses of the endmember hydrothermal fluids reveal that the Aurora system discharges highly acidic, silicon-poor fluids. Geologically comparable ultraslow-spreading Arctic systems like Loki's Castle reliably measure between 5.50 and 6.06 (Pedersen et al., 2010). Other sources indicate that similar mafic-ultramafic sites, such as the Von Damm field, also feature an inferred pH of approximately 5.5 to 6, supporting a moderately acidic profile (Albers et al., 2025). The literature exhibits no fundamental disagreement regarding the broadly acidic nature of these hydrothermal fluids, though the lack of published numerical data for Aurora itself requires reliance on these regional analogues.

Reliability: 1 (Low). This score is assigned because exact numerical pH data for Aurora remains unpublished, requiring reliance on proxy measurements from comparable ultraslow-spreading hydrothermal systems.

c. Salinity (g/kg): ~34.9
Source: Oceanographic profiles establish that the background deep seawater surrounding and mixing with the Aurora hydrothermal plume maintains a practical salinity of approximately 34.91 to 34.93 psu, which translates to roughly 34.9 g/kg (Walter et al., 2025). Other sources corroborate this by describing the deep Arctic Ocean as weakly stratified and dominated by a consistent saline environment. No disagreement exists regarding the ambient salinity, although the specific undiluted salinity of the hydrothermal endmember fluid has not been explicitly decoupled from these deep-water measurements in the available texts (Mertens, 2025, Walter et al., 2025).

Reliability: 1 (Low). Data does not specify salinity of the fluids themselves, and is instead derived from in-situ conductivity-temperature-depth (CTD) sensor measurements for surrounding ambient seawater.

d. Pressure (MPa): ~39 to 40
Source: Peer-reviewed bathymetric and remotely operated vehicle (ROV) data conclusively place the active Aurora vents at water depths ranging from 3880 metres to roughly 4000 metres (Jakuba et al., 2024, Ramirez-Llodra et al., 2023). This depth range corresponds to a hydrostatic pressure of approximately 39 to 40 MPa. All scientific expeditions agree on this depth and subsequent pressure range, as it has been consistently verified by multiple sonar, bathymetric, and ROV-mounted depth sensors across different cruise missions without any contention (Bünz et al., 2021, Lapointe et al., 2025).

Reliability: 3 (High). Based on primary bathymetric mapping and direct ROV telemetry data published in peer-reviewed oceanographic journals.

e. Isolation Score (1-5): 2
(i) Physical Confinement (1-5): 2
Source: The vent field is situated in the deep Arctic Ocean under a perennial sea-ice cover, providing partial confinement from the atmosphere while remaining an open ocean system (Ramirez-Llodra et al., 2023, Walter et al., 2025).

(ii) Hydrologic Connectivity (1-5): 2
Source: The Arctic Ocean is bathymetrically isolated from other oceans by shallow sills, meaning the deep waters surrounding the vents connect only to a few adjacent bodies via the Fram Strait (Walter et al., 2025).

(iii) Isolation Time (1-5): 2
Source: Despite permanent ice cover, the deep water masses of the Eurasian Basin experience renewal and mixing on timescales of hundreds of years, allowing for regular, albeit slow, regional exchange (Björk & Winsor, 2006, cited in Walter et al., 2025).

Reliability: 3 (High). This score reflects that the isolation parameters are derived from peer-reviewed physical oceanography studies and direct hydrographic measurements.

f. Redox Score (1-5): 5
(i) Redox Species (1-5): 5
Source: Primary fluid sampling proves the presence of extremely high levels of reduced species, including molecular hydrogen approaching 50 mmol/L and methane exceeding 1 mmol/L, venting into oxygenated deep-sea water (German et al., 2022, Wegener et al., 2023).

(ii) Equilibrium (1-5): 5
Source: A highly exploitable redox gradient is created when the extremely H2-rich, hot, and acidic hydrothermal fluids mix directly with the cold, well-oxygenated background seawater of the Arctic Ocean (Lapointe et al., 2025, Walter et al., 2025).

(iii) Chemolithotrophic Metabolism (1-5): 5
Source: There is strong, measured evidence of chemolithotrophic metabolism in the vent plume, primarily driven by hydrogen-oxidizing bacteria like Candidatus Sulfurimonas pluma fixing carbon at high rates (Wegener et al., 2023).

Reliability: 3 (High). This score is justified because the redox and metabolic data come from direct in-situ chemical sampling and genetic sequencing published in peer reviewed journals.

13. Lake Whillans, Antarctica

Scope Definition: The site encompasses the subglacial water column (approximately 2.2 m deep) and the surficial interacting basal sediments (top ~40 cm) located beneath 801 m of glacial ice on the Whillans Ice Stream in West Antarctica.

a. Temperature (°C): -0.5
   Source: In situ measurements from the conductivity-temperature-depth profiler deployed by Tulaczyk et al. (2014) identify the water column temperature as precisely -0.5°C. Other synthesis papers confirm that this temperature accurately reflects the pressure-melting point at the base of the approximately 800-metre-thick West Antarctic Ice Sheet (Christner et al., 2014; Mikucki et al., 2016). There is no disagreement among the sources, as all researchers rely on the definitive sensor string data collected during the direct clean-access drilling campaign in 2013.

   Reliability: 3 (High). The data originates from in situ field measurements using calibrated sensors deployed through a pristine access borehole.

b. pH: 8.1
   Source: Geochemical measurements report the bulk water column pH of Subglacial Lake Whillans as 8.1 (Michaud et al., 2020; Christner et al., 2014). Other studies focusing on the deeper porewaters show highly variable pH estimates due to localised sediment reactions, but all researchers accept the mildly alkaline 8.1 value for the primary lake water (Mikucki et al., 2016). There is no significant disagreement regarding this water column pH, which is heavily influenced by silicate weathering processes.

   Reliability: 3 (High). Direct, peer-reviewed collection and analysis of pristine water samples utilizing established, clean-access field measurement protocols.

c. Salinity (g/kg): 0.55 to 0.65
   Source: Thermodynamic recalculations of original physical data using modern equations of state determine the absolute salinity of the subglacial water column to be approximately 0.55 to 0.65 g/kg (Mikucki et al., 2016). Initial field measurements originally inferred practical salinity to be between 0.37 and 0.41 practical salinity units, while other geochemical studies simply report the dilute electrical conductivity of 720 µS/cm (Tulaczyk et al., 2014, Christner et al., 2014). Disagreement stems purely from the evolution of limnological calculation methods, with updated frameworks producing absolute salinity values roughly fifty percent higher than the initially inferred practical measurements (Mikucki et al., 2016).

   Reliability: 3 (High). Conductivity and salinity data are derived from highly controlled laboratory analyses of cleanly extracted lake water.

d. Pressure (MPa): ~7.1 to 8.0
Source: The best available glaciological sources document that the lake is situated beneath exactly 801 metres of glacial ice, creating an immense hydrostatic and lithostatic pressure environment (Tulaczyk et al., 2014; Michaud et al., 2016). While the sources do not explicitly state the precise pressure in megapascals, standard physical conversions for 800 metres of ice overburden reliably place the basal pressure at approximately 7.1 to 8.0 MPa (Mikucki et al., 2016). No sources contest this massive depth or the resultant high-pressure subglacial conditions.

Reliability: 3 (High). Pressure is directly extrapolated from precisely measured borehole depth surveys obtained during the 2013 drilling campaign.

e. Isolation Score (1-5):
(i) Physical Confinement (1-5): 5
Source: The lake is physically confined beneath 801 metres of the West Antarctic Ice Sheet, permanently preventing any direct atmospheric or sunlight exposure (Tulaczyk et al., 2014). This thick, permanent ice cover enforces a strictly dark, isolated subglacial habitat.

(ii) Hydrologic Connectivity (1-5): 3
Source: Subglacial Lake Whillans acts as an active, integrated part of a regional subglacial drainage network (Fricker et al., 2007). It periodically receives water from upstream lakes and flushes into the downstream marine cavity.

(iii) Isolation Time (1-5): 2
Source: Because the lake experiences periodic, sub-decadal fill-and-drain cycles, the actual water residence time is relatively short (Siegfried et al., 2014). Consequently, the fluid remains isolated in the lake basin for only years to decades (Michaud et al., 2016).

Reliability: 3 (High). Firmly supported by robust satellite altimetry data and direct geochemical validations.

f. Redox Score (1-5):
(i) Redox Species (1-5): 4
Source: The subglacial environment features a prominent redox gradient where oxygenated glacial meltwater interfaces with ancient, anoxic marine sediments (Michaud et al., 2016). This supplies distinct, exploitable oxidants like oxygen and reductants like methane.

(ii) Equilibrium (1-5): 4
Source: A strong, exploitable thermodynamic gradient exists due to the disequilibrium between the mildly oxic water column and the highly reduced sediments (Mikucki et al., 2016). Microbial activity actively exploits these gradients, preventing chemical equilibrium.

(iii) Chemolithotrophic Metabolism (1-5): 5
Source: There is confirmed evidence of a thriving microbial ecosystem heavily dominated by chemolithoautotrophic bacteria and archaea (Christner et al., 2014). Genomic data prove these organisms actively metabolize reduced iron, sulfur, and nitrogen compounds for energy.

Reliability: 3 (High) Validated by direct genomic sequencing, thermodynamic modelling, and in-situ biogeochemical assays.

14. Lake Vida, Antarctica

Scope Definition: The interstitial, aphotic, and anoxic liquid brine network located beneath 16 to 27+ meters of perennial ice cover in Lake Vida, Victoria Valley, Antarctica, excluding the surface ice and peripheral meltwater.

a. Temperature (°C): -13.4 to -11.6
   Source: Doran et al. (2003) and Malone et al. (2010) say the temperature is -11.6 °C because their measurements were based on early ice core data extracted in 1996 where briny ice was initially encountered at 15.8 m depth. Murray et al. (2012) and Dugan et al. (2015) say the temperature is -13.4 °C because their data reflects direct, in-situ measurements of the liquid brine after it infiltrated boreholes drilled during the 2005 and 2010 expeditions. The union range of -13.4 °C to -11.6 °C is adopted because it captures both the early ice matrix thermal profiles and the later confirmed liquid brine measurements, reflecting slight potential variations in the subsurface network.

   Reliability: 3 (High). The data is derived from direct in-situ field measurements using calibrated sensors documented in peer-reviewed scientific journals.

b. pH: 6.2
   Source: Sources say that the pH of the Lake Vida brine is exactly 6.2, characterizing it as a slightly acidic environment (Murray et al., 2012). Other sources land on this exact value, consistently confirming the 6.2 measurement across multiple sampling years and independent studies (Kuhn et al., 2014; Ostrom et al., 2016). There is no disagreement regarding the pH of the brine, as the 2005 and 2010 expedition samples yielded identical geochemical characterizations regarding its slight acidity in the anoxic subsurface.

   Reliability: 3 (High). The parameter was directly measured from pristine samples and published in peer-reviewed literature without any conflicting reports.

c. Salinity (g/kg): 175 to 245
Source: Doran et al. (2003) say the salinity is 245 g/L because they initially inferred the concentration using freezing-point depression thermodynamics at the deep ice-brine interface. Malone et al. (2010) and Murray et al. (2012) say the salinity is between 175 and 210 g/L because subsequent geochemical analysis, refractive index measurements, and noble gas modeling on extracted liquid brine samples indicated a lower salt concentration. The union range of 175 to 245 g/L is adopted because it accounts for both the localized cryoconcentration estimates in the ice matrix and the direct bulk measurements of the liquid brine network.

Reliability: 3 (High). These values emerge from direct field sampling, rigorous laboratory geochemical analyses, and peer-reviewed primary mission data.

d. Pressure (MPa): ~0.11
Source: Sources say that the brine exists under elevated hydrostatic pressure, as evidenced by subsurface brine rising from 16 m to 10.5 m in boreholes (Dugan et al., 2015). Other sources land on a quantitative estimate of 1.1 atm (which converts to 0.11 MPa) based on noble gas phase-partitioning models matching current brine sample conditions (Malone et al., 2010). The nature of any disagreement is minimal, but the exact in-situ numerical pressure is rarely explicitly defined in MPa, leaving it derived from physical modeling and observation.

Reliability: 2 (Medium). This score is assigned because while the pressurized nature is directly observed, the specific numerical pressure values are derived from modelling rather than direct sensors.

e. Isolation Score (1-5):
(i) Physical Confinement (1-5): 5
Source: Dugan et al. (2015) demonstrate that the lake features an exceptionally thick perennial ice cover exceeding 27 meters. This physical barrier perfectly aligns with the "thick ice cover" criteria, effectively sealing the liquid from surface interactions.

(ii) Hydrologic Connectivity (1-5): 5
Source: Dugan et al. (2015) confirm the brine system acts as a confined aquifer without a valley-wide groundwater network. The complete absence of atmospheric or peripheral liquid exchange categorizes this as a highly sealed system with no exchange.

(iii) Isolation Time (1-5): 3
Source: Doran et al. (2003) and Murray et al. (2012) provide radiocarbon dating of organic matter indicating isolation for at least 2,800 to 4,000 years. This duration places the site comfortably in the 1,000 to 100,000 years isolation bracket.

Reliability: 3 (High). This score is awarded because the isolation parameters are supported by rigorous peer-reviewed geophysical imaging and radiocarbon dating of the subsurface environment.

f. Redox Score (1-5):
(i) Redox Species (1-5): 5
Source: Murray et al. (2012) confirm the presence of high concentrations of both molecular hydrogen (10.5 umol/L) and carbon dioxide (8.9 mmol/L). This measured evidence of both species perfectly satisfies the criteria for the highest possible score.

(ii) Equilibrium (1-5): 4
Source: Pester et al. (2018) highlight a severe isotopic disequilibrium between molecular hydrogen and water. This profound disequilibrium, alongside co-occurring oxidized and reduced nitrogen species, provides strong evidence for an exploitable thermodynamic gradient (Murray et al., 2012).

(iii) Chemolithotrophic Metabolism (1-5): 4
Source: Murray et al. (2012) report extremely low rates of protein synthesis and limited metabolic activity. While chemolithoautotrophic bacteria are present, the extreme cold and high salinity provide only weak, maintenance-level evidence of active chemolithotrophic metabolism in situ.

Reliability: 3 (High). This score is justified because the redox state, chemical species, and biological activity were measured directly via peer-reviewed geochemical assays and genomic sequencing.

15. Don Juan Pond, Antarctica

Scope Definition: A shallow, hypersaline calcium chloride pond and its immediately associated surface brine and localized groundwater network situated in the closed-basin South Fork of Wright Valley, Antarctica.

a. Temperature (°C): -50 to 10
   Source: High-quality sources agree that Don Juan Pond experiences extreme temperature variations, with surface temperatures ranging from -50°C during the austral winter to +10°C in the summer (Marion, 1997). Thermodynamic models and field measurements concur that the high calcium chloride concentration depresses the freezing point to a eutectic temperature of roughly -51.8°C, preventing the pond from freezing completely (Toner et al., 2017). There is virtually no disagreement among researchers regarding these temperature dynamics, as sustained meteorological monitoring confirms this environmental parameter (Gough et al., 2017).

   Reliability: 3 (High). Temperatures are derived from direct field measurements with calibrated meteorological stations and supported by robust thermodynamic phase models (Toner et al., 2017).

b. pH: 5.57
Source: Peer-reviewed geochemical analysis of Don Juan Pond reports an acidic pH value of 5.57 for the surface brine (Webster, 1994). Other literature largely references this baseline chemistry or focuses entirely on the extreme salinity and ionic ratios rather than pH fluctuations. There is no significant disagreement regarding the pond's acidity, though it is acknowledged that sampling is infrequent and seasonal variations could hypothetically exist due to minor meltwater influxes.

Reliability: 3 (High). The pH data originates from peer-reviewed primary field sampling utilizing calibrated field electrodes lowered through the pond's profile (Webster, 1994).

c. Salinity (g/kg): 400 to 442
Source: Investigations report that Don Juan Pond is the most saline natural water body on Earth, with a salinity exceeding 400 g/kg or 40% by mass (Dickson et al., 2013). Other sources completely align with this, noting the brine is a dense, near-saturated calcium chloride solution composed of 413 g/kg CaCl2 and 29 g/kg NaCl (Samarkin et al., 2010). There is no disagreement regarding its hyper-salinity, though the exact concentration fluctuates slightly depending on seasonal evaporation and shallow groundwater recharge rates (Toner et al., 2017).

Reliability: 3 (High). These figures are backed by peer-reviewed literature and decades of direct field sampling and chemical analyses (Samarkin et al., 2010; Toner et al., 2017).

d. Pressure (MPa): ~0.1
Source: The Don Juan Pond is a subaerial surface feature situated at an elevation of 117 to 122 metres above sea level (Toner et al., 2017; Marion, 1997). Consequently, the pond and its immediate near-surface groundwater interface operate at standard atmospheric pressure, approximately 0.1 MPa. The pond is an open-air, extremely shallow pond averaging only 10 centimetres in depth. There is absolutely no disagreement regarding the ambient pressure conditions of this subaerial environment.

Reliability: Score 3. Pressure is easily derived from the confirmed surface elevation recorded by peer-reviewed geological and geographic surveys (Marion, 1997).

e. Isolation Score (1-5): 2
(i) Physical Confinement (1-5): 1
Source: Due to its extreme calcium chloride salinity, Don Juan Pond remains an open water body and rarely freezes, even at temperatures reaching -50°C (Dickson et al., 2013). This completely limits any physical confinement, leaving the brine fully exposed to the atmosphere.

(ii) Hydrologic Connectivity (1-5): 2
Source: Group A says the pond is fed by an expansive deep regional groundwater aquifer because of consistent ionic ratios and upwelling pressures observed in local boreholes (Toner et al., 2017; Dugan et al., 2022). Group B says the pond is fed primarily by near-surface, shallow water tracks and deliquescent salts that flow laterally over the local permafrost (Dickson et al., 2013; Gough et al., 2017).

(iii) Isolation Time (1-5): 1
Source: Geochemical modelling indicates that the pond operates as an open, flow-through system where evaporated brines continuously recycle back into the subsurface. This keeps the residence time of surface waters to less than a single year (Toner et al., 2017).

Reliability: 3 (High). Isolation metrics are supported by peer-reviewed hydrogeological surveys, active time-lapse monitoring, and robust geochemical mixing models (Toner et al., 2017; Dickson et al., 2013).

f. Redox Score (1-5): 3
(i) Redox Species (1-5): 4
Source: The pond exhibits elevated concentrations of molecular hydrogen, exceeding atmospheric levels by a factor of 700, alongside abundant nitrate, nitrite, and Fe(II)-rich dolerite minerals (Samarkin et al., 2010). This represents strong evidence of highly reactive reduced and oxidized species.

(ii) Equilibrium (1-5): 4
Source: A strong, exploitable thermodynamic gradient exists as the hypersaline brine reacts abiotically with surrounding dolerite. This reaction drives a continuous and measurable flux of nitrous oxide and hydrogen gas into the atmosphere via chemodenitrification (Samarkin et al., 2010).

(iii) Chemolithotrophic Metabolism (1-5): 1
Source: Despite the presence of a steep redox gradient, all attempts to measure active biological activity or microbial metabolism within the extreme hypersaline brine itself have consistently failed (Samarkin et al., 2010; Toner et al., 2017).

Reliability: 3 (High). Redox scores are derived from peer-reviewed in situ flux measurements and controlled laboratory chemodenitrification experiments (Samarkin et al., 2010).

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